

Quiz 8: Fibers & Chromatography

Review of Session 8. ~15 minutes.

Section A: Fiber burn test

For each described result, name the class (animal / vegetable / synthetic) and a likely specific fiber:

1. Curls away, burns slowly with sputtering, self-extinguishes, smells like burning hair, black crushable bead.
2. Ignites readily, burns steadily with an afterglow, smells like burning paper, light gray ash, no bead.
3. Shrinks and melts away from the flame, drips, hard shiny bead, sharp chemical smell.
4. Burns very fast with almost no melting, smells like paper, little ash.

Section B: Fiber facts

1. Most synthetic fibers are made from what raw material?
2. Rayon and acetate are made by regenerating what natural material?
3. What kind of fiber is best for moisture-wicking athletic wear, and how does the wicking work?
4. What is the difference between direct and secondary fiber transfer?

Section C: Chromatography

1. $R_f = \frac{\text{distance of spot}}{\text{distance of solvent front}}$
2. What is the range of possible R_f values?
3. What is the stationary phase in paper chromatography? The mobile phase?
4. Why must the start line be drawn in pencil, not pen?
5. A dye spot moved 3 cm; the solvent front moved 12 cm. What is the R_f ?
6. Who is credited with inventing chromatography, and what does the word mean?
7. To match a crime-scene ink to a suspect's pen, what do you compare?

